

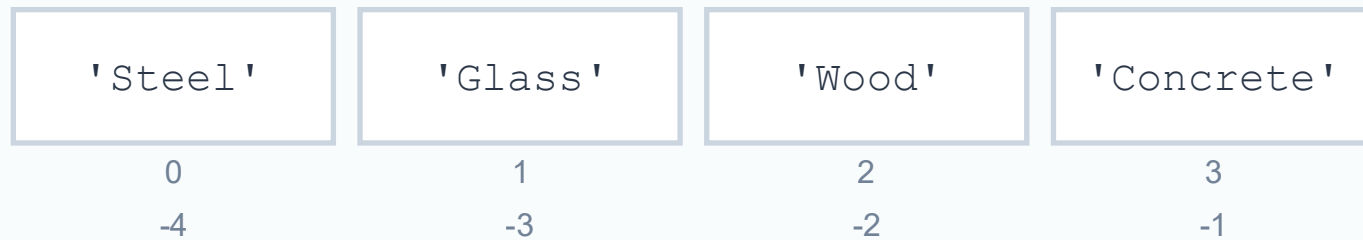
Python List Operations

Indexing, Slicing, and Modifying Ordered Collections

1. Indexing (Accessing Items)

Access single items using their position (index).

```
materials = ['Steel', 'Glass', 'Wood', 'Concrete']
```



```
print(materials[2]) # Output: Wood
```

2. Slicing (Accessing Sub-Lists)

Extract a range of items. Format: `[start:stop]`.

```
materials = ['Steel', 'Glass', 'Wood', 'Concrete']
```



```
facade = materials[1:3] # Gets ['Glass', 'Wood']
```

3. Adding Items

`.append()` adds to the end. `.insert()` adds at an index.

```
.append('Brick')
```

Before: ['Steel', 'Glass']

After: ['Steel', 'Glass', 'Brick']

```
.insert(1, 'Stone')
```

Before: ['Steel', 'Glass']

After: ['Steel', 'Stone', 'Glass']

```
materials.append('Brick')
materials.insert(1, 'Stone')
```

4. Removing Items

`.pop()` removes by index. `.remove()` removes by value.

```
.pop(0)
```

Before: ['Steel', 'Glass', 'Wood']

After: ['Glass', 'Wood'] ('Steel' was removed)

```
.remove('Glass')
```

Before: ['Steel', 'Glass', 'Wood']

After: ['Steel', 'Wood'] ('Glass' was removed)

```
materials.pop(0)
materials.remove('Glass')
```